

10) Let $F(x, y)$ be the statement " x can fool y ," where the domain consists of all people in the world. Use quantifiers to express each of these statements.

- a) Everybody can fool Fred: $\forall x F(x, \text{Fred})$
- b) Evelyn can fool everybody: $\forall x F(\text{Evelyn}, x)$
- c) Everybody can fool somebody: $\forall x \exists y F(x, y)$
- d) There is no one who can fool everybody: $\forall x \exists y \neg F(x, y)$
- e) Everyone can be fooled by somebody: $\forall y \exists x F(x, y)$
- f) No one can fool both Fred and Jerry: $\forall x (\neg (F(x, \text{Fred}) \wedge F(x, \text{Jerry})))$

- g) Nancy can fool exactly two people: $\exists x \exists y (x \neq y) \wedge F(\text{Nancy}, x) \wedge F(\text{Nancy}, y) \wedge \forall z (F(\text{Nancy}, z) \Rightarrow (z = x \vee z = y))$

- h) There is exactly one person whom everybody can fool: $\exists x \exists y \forall z (F(z, x) \wedge F(z, y) \Rightarrow (x = y))$

- i) No one can fool himself or herself: $\forall x \neg F(x, x)$
- j) There is someone who can fool exactly one person besides himself or herself: $\exists x (F(x, x) \wedge \exists y \exists z ((x \neq y) \wedge (x \neq z) \wedge F(x, y) \wedge F(x, z) \Rightarrow (y = z)))$

11) Let $S(x)$ be the predicate " x is a student," $F(x)$ the predicate " x is a faculty member," and $A(x, y)$ the predicate " x has asked y a question," where the domain consists of all people associated with your school. Use quantifiers to express each of these statements.

- a) Lois has asked Professor Michaels a question: $A(\text{Lois}, \text{Professor Michaels})$
- b) Every student has asked Professor Miller a question: $\forall x (S(x) \Rightarrow A(x, \text{Professor Miller}))$
- c) Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller: $\forall x (F(x) \Rightarrow (A(x, \text{Professor Miller}) \vee A(\text{Professor Miller}, x)))$
- d) Some student has not asked any faculty member a question: $\neg \forall x (S(x) \wedge \forall y (F(y) \Rightarrow \neg A(x, y)))$